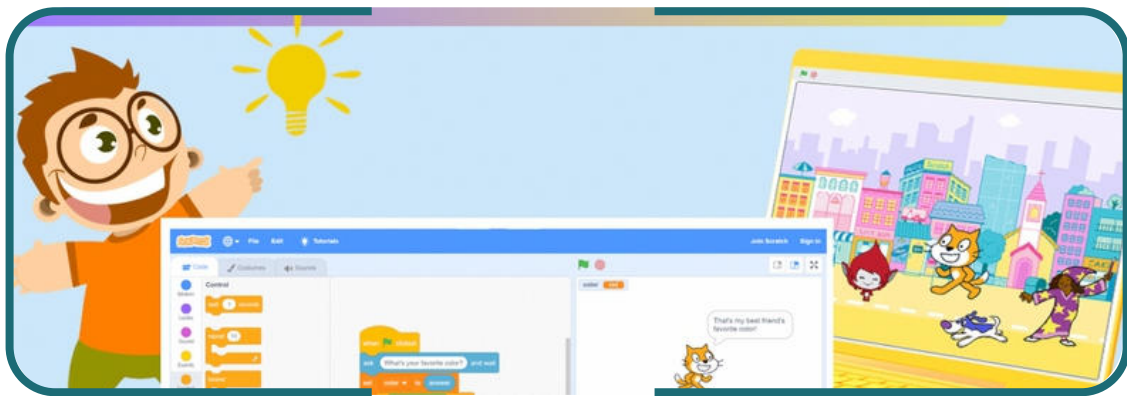


UNIT 8

GETTING CREATIVE WITH SCRATCH – LOOPS, VARIABLES, AND MORE!



Learning Objectives

- Identify and describe the parts of the Scratch interface.
- Create simple animations or games using Scratch blocks.
- Use motion, looks, sound, control, and event blocks effectively in Scratch.
- Understand how loops help repeat actions.
- Learn about variables and how to use them in Scratch.
- Apply loops (repeat, forever) to automate actions in Scratch.
- Create and manipulate variables to store information in Scratch.
- Use Scratch loops and variables to draw geometric shapes.
- Add sound effects and backgrounds to enhance animations in Scratch.
- Discover conditional statements for creating interactive projects.

Scratch is a block-based visual programming language designed for beginners, especially kids. It enables users to create simple games and animations by dragging and dropping blocks instead of writing text-based code. The blocks snap together, allowing users to control sprites (characters or objects) and define their actions in a simple, intuitive way.

Parts of the Scratch Screen:

- **Stage:** The main area where you view your animations and sprite actions, like a theatre stage.
- **Sprite Pane:** Displays all the sprites in your project. Each sprite acts as a character or object.
- **Block Palette:** Contains different blocks categorized by function (e.g., Motion, Looks, Sound).
- **Script Area:** Workspace where you build scripts by snapping blocks together to give sprites instructions.
- **Backdrop:** The background of the stage, which sets the scene for your animation or game.

1. Adding Backgrounds and Sounds

Changing the Backdrop:

1. In the Stage area, click on the "Choose a Backdrop" button.
2. Select a background, like a beach or a city. The backdrop will change on the stage.

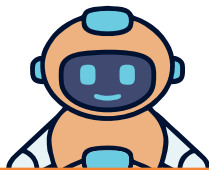
Adding Sound:

1. Go to the Sound category in the Block Palette.
- Drag the "play sound Meow until done" block to your script.
- Choose from different sounds or record your own!

2. Scratch Blocks

- **Motion Blocks:** Control sprite movement (e.g., "move 10 steps," "turn 15 degrees").
- **Looks Blocks:** Change sprite appearance (e.g., "say [message] for 2 seconds," "change costume").
- **Sound Blocks:** Add sounds to your project.
- **Control Blocks:** Manage timing and loops (e.g., "wait 1 second," "repeat 10 times").
- **Events Blocks:** Start actions, like "when green flag clicked."

01: Lab Activity : Create an Animated Scene with Sound



Create an animation using movement, sounds, and changing backgrounds in Scratch.

Instructions:

1. Choose a Sprite and Background

- Select a sprite (e.g., a cat) and a starting backdrop (e.g., city).

1. Add Movement

- Go to Motion blocks and add "move 10 steps."
- Wrap it in a "repeat 10" block from Control to make the sprite move multiple times.

2. Add Sound

- Go to Sound blocks, drag "play sound Meow until done," and place it in the loop.

3. Change Background

- After the loop, add "switch backdrop to" and select a new backdrop.

4. Run the Animation

- Click the green flag to see your sprite move, play sound, and change the backdrop.

5. Loops: Making Actions Repeat

Loops allow actions to repeat without adding the same block multiple times.

- **Repeat Loop:** Repeats an action a specific number of times (e.g., "repeat 10").

- **Forever Loop:** Keeps repeating until you stop it, perfect for ongoing animations.



6. Variables: Storing Information

Variables act like containers for information that can change, such as score or player name, adding interactivity.



Creating a Variable:

1. Go to the Variables category.
2. Click "Make a Variable."
3. Name your variable (e.g., "Score").

How to Draw Shapes Using Variables and Loops

1. Create Variables:

- Go to the "Variables" category and create two variables:
- Number of Sides
- Side Length

2. Set Up Your Variables:

- Use the set [variable] to [value] block to assign initial values:
- Set Number of Sides to the desired number of sides (e.g., 4 for a square, 3 for a triangle).
- Set Side Length to the length of each side.

3. Use a Loop to Draw the Shape:

- Use a repeat [] block to repeat actions for the number of sides:
- Inside the loop:
- Add a move [Side Length] steps block to draw one side.
- Add a turn $[360 \div \text{Number of Sides}]$ degrees block to turn the sprite correctly for the next side.

4. Calculate the Angle Dynamically:

- Use a division operator from the "Operators" category to calculate the turn angle: $360 \div \text{Number of Sides}$.

Example Script to Draw a Shape:

Here's how the Scratch script would look:



plaintext

 Copy code

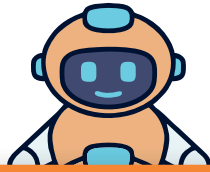
```
when green flag clicked
set [Number of Sides] to [4] // Change to desired number of sides
set [Side Length] to [50]    // Change to desired side length
clear                        // Clears the drawing area (optional)
pen down                     // Starts drawing
repeat (Number of Sides)
  move (Side Length) steps
  turn (360 / Number of Sides) degrees
end
pen up                       // Stops drawing (optional)
```

DO YOU KNOW?

Variables are temporary memory locations in the computer while you run a program. During execution of the program, the values are temporarily stored in the variables while

Loops are used to repeat the code.

Lab Activity 1: Draw a Shape Based on User Input

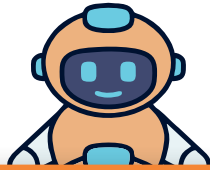


Create a program to draw any shape by asking the user for the number of sides and side length.

Instructions:

1. Open Scratch and create a new project.
2. Add two variables: Number of Sides and Side Length.
3. Use the ask [] and wait block to:
 - Ask the user: "How many sides does the shape have?" and set Number of Sides to the answer.
 - Ask: "What is the side length?" and set Side Length to the answer.
4. Use a repeat [Number of Sides] block to:
 - Move the sprite forward by Side Length steps.
 - Turn the sprite by $360 \div \text{Number of Sides}$ degrees.
5. Add pen down at the start and pen up at the end to draw the shape.
6. Click the green flag and test the program with different inputs!

Lab Activity 2: Create a Pattern with Shapes



Use a loop to draw a pattern
made of multiple shapes

Instructions:

1. Open a new or existing project from Activity 1.
2. Add a variable: Number of Shapes.
3. Set Number of Shapes to the number of times you want the shape repeated in a pattern.
4. Wrap the shape-drawing loop from Activity 1 inside another repeat [Number of Shapes] loop.
5. After each shape is drawn, add a turn [15] degrees block to rotate the sprite slightly.
6. Use the set pen color to [pick random 1 to 200] block before each shape to create colorful patterns.
7. Test your program with different numbers of sides, side lengths, and pattern repetitions.

Conditional Statements: Making Decisions

Conditionals allow scripts to make decisions based on certain conditions using "if" or "if else" blocks, adding interactivity.

Example: In a game, if a sprite touches a red area, it stops and says, "Game Over!" using the "if touching colour" block.

Combining Loops, Variables, and Conditionals

By combining loops, variables, and conditionals, you can create complex projects. For example, in a maze game, you can:

- Use a variable to track score.
- Add a loop for continuous sprite movement.
- Use an "if" block to check if the sprite reaches the goal.

Example Project: Collecting Coins Game

Create a game where a sprite collects coins:

1. Add Sprites: Choose a main character and a coin sprite.
2. Movement: Use arrow keys to move your main character.
3. Collect Coins: Each time the main sprite touches a coin, it "collects" it and the coin moves to a new spot.
4. Track Score: Use a variable to increase the score by 1 each time the main sprite collects a coin.
5. End the Game: Set a goal score, like 10, and display a message such as "You Win!" when reached.

EXERCISES

01: Choose the correct answer.

1.What is the primary function of the "Motion" blocks in Scratch?

- (a) Control sprite movement
- (b) Change the sprite's color
- (c) Add sound effects

2.Which of the following is used to store dynamic information like the score in a Scratch project?

- (a) Sound blocks
- (b) Variables
- (c) Backdrops

3.In Scratch, what happens when you use a "repeat" block?

- (a) It repeats a block of code a specified number of times
- (b) It stops the program after a set number of actions
- (c) It turns the sprite to a random position

4.How can you change the background in a Scratch project?

- (a) Using the "Looks" block
- (b) By clicking the "Choose a Backdrop" button in the Stage area
- (c) By adding a new sprite

5.What does the "Forever" loop in Scratch do?

- (a) Repeats an action continuously until stopped
- (b) Repeats an action a specified number of times
- (c) Executes a set of actions only once

6.Which block category in Scratch allows you to change the appearance of a sprite?

- (a) Motion
- (b) Looks
- (c) Control

7.What is the purpose of the "pen down" block in Scratch?

- (a) It starts drawing a shape
- (b) It stops drawing a shape
- (c) It plays a sound effect

8.How do you calculate the turn angle for drawing a shape in Scratch?

- (a) Divide 360 by the number of sides of the shape
- (b) Multiply 360 by the number of sides of the shape
- (c) Add 360 to the number of sides

9.Which block is used to make a sound play in Scratch?

- (a) "play sound Meow until done"
- (b) "move 10 steps"
- (c) "say [message] for 2 seconds"

10.In Scratch, which category would you use to create a new variable for storing score information?

- (a) Motion
- (b) Control
- (c) Variables

11.Which type of loop keeps repeating forever until you stop it?

- (a) Repeat Loop
- (b) Forever Loop
- (c) Conditional Loop

12.What is a variable used for in Scratch?

- (a) Moving a sprite
- (b) Storing information that can change
- (c) Adding sounds to a project

13. Which block would you use to check if a sprite is touching a certain color?

- (a) If touching colour
- (b) Repeat until touching colour
- (c) Forever touching colour

14. What would you use to keep track of a score in a Scratch game?

- (a) A loop
- (b) A variable
- (c) A costume change

15. Which of the following blocks would you use to repeat an action 10 times?

- (a) Forever
- (b) Repeat 10
- (c) Wait 10 seconds

02: Fill in the blanks.

1. A _____ allows us to repeat actions without adding the same block multiple times.
2. The _____ loop in Scratch continues until it's stopped manually.
3. _____ are used to store information, like a player's score or name.
4. The _____ block lets us check if a condition is true or not.
5. _____ statements make projects interactive by allowing Scratch to make decisions.

03: Tick True or False for each statement.

	TRUE	FALSE
A variable can store text, such as a player's name, or a number, like a score.	☐	☐
The "Repeat 10" loop will keep repeating forever.	☐	☐
Conditionals in Scratch allow you to create simple animations but not games.	☐	☐
Loops help make scripts shorter and easier to read.	☐	☐
The "If touching colour" block is found in the Variables category	☐	☐

04: Match the terms with their correct definitions.

Column A	Column B
1. Repeat 10	a. Keep score in a game
2. Variable	b. Checks if something is true
3. Forever Loop	c. Repeats an action 10 times
4. If Block	d. Stores information
5. Conditionals	e. Repeats an action until stopped

05: Answer the following questions briefly.

1. What is the difference between a Repeat loop and a Forever loop?
2. Give an example of a situation in Scratch where you might use a variable.
3. What are conditional statements used for in Scratch?
4. How does a variable help in making a game interactive?
5. Why are loops helpful when creating animations?

06: Provide detailed answers for the following questions.

- 1.** Explain how the "repeat" and "forever" loops differ in Scratch and provide examples of how each can be used in a project.
- 2.** Describe how you would use variables in Scratch to store and update a score in a game. Provide an example script.
- 3.** Explain the process of creating an animation in Scratch, including adding movement, sound, and changing backgrounds.
- 4.** Describe how to create a polygon in Scratch using loops and variables for the number of sides and side length. Include the necessary steps and blocks.

Lab Activity 3: Create a Moving Ball Game



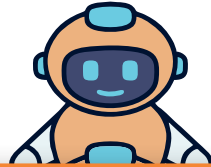
Create a game where a ball moves around the stage and stops if it touches the red colour.

Instructions:

1. Add a ball sprite and draw a red line on the backdrop as a boundary.
2. Use a Forever loop to keep the ball moving across the stage.
3. Add an "If touching colour red" block to stop the ball if it touches the red line.

Extension: Add a variable to count how many times the ball reaches the edge of the stage.

Lab Activity 4: Make a Score Tracker



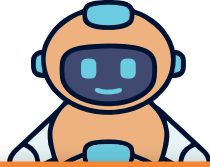
Use a variable to keep track of the score
each time a sprite is clicked.

Instructions:

1. Choose a sprite, like a balloon.
2. Create a variable called "Score."
3. Create a script so that each time the balloon is clicked, it moves to a new spot, and the score increases by 1.

Extension: End the game when the score reaches 10 and display a "You Win!" message.

Lab Activity 5: Simple Quiz Game



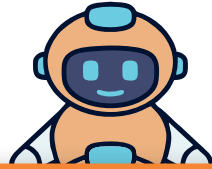
Create an interactive quiz with questions that use conditionals to check the answers.

Instructions:

1. Ask a question, like "Is the sky blue?" and use the "Ask [question] and wait" block.
2. Use an "If" block to check if the answer is "yes."
3. If the answer is correct, increase the score by 1.

Extension: Add more questions and keep track of the score using a variable.

Lab Activity 6: Animate a Sprite with Loops



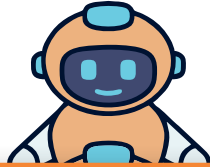
Use a repeat loop to create a simple animation where a sprite "dances."

Instructions:

1. Choose a sprite with multiple costumes.
2. Use a Repeat loop to switch between costumes and create a dance move.
3. Add a sound effect to make the dance more fun.

Extension: Add a variable to keep track of how many dance moves have been completed.

Lab Activity 7: Create a Countdown Timer



Use a variable to create a countdown timer for a game.

Instructions:

1. Create a variable called "Timer" and set it to 10.
2. Use a loop to decrease the timer by 1 every second.
3. Add an "if Timer = 0" block to end the game when the countdown reaches zero.

Extension: Display a "Game Over!" message when the timer hits zero.